In [2]:
```
import pandas as pd
import numpy as np
```

In [3]:
```
df=pd.read_csv("1. hr_analytics.xls")
print(df)
```

```
     Age  Gender Department  MonthlyIncome  TotalWorkingYears  YearsAtCompany  \
0     50  Female         HR           4753                 14               8
1     36    Male        R&D          13123                  3               7
2     29    Male         HR          19376                 16              15
3     42    Male         HR           9588                 19              14
4     40    Male        R&D           9304                 18               8
..   ...     ...        ...            ...                ...             ...
995   34  Female      Sales           7986                  2               5
996   51  Female      Sales           4481                 26              12
997   44    Male         HR           6731                 27               6
998   40  Female      Sales          19835                 27               9
999   53  Female         HR          13758                 15               1

     JobSatisfaction  WorkLifeBalance OverTime Attrition
0                  2                1      Yes        No
1                  3                4       No        No
2                  1                3      Yes        No
3                  2                4       No       Yes
4                  2                3       No       Yes
..               ...              ...      ...       ...
995                4                3      Yes       Yes
996                4                2      Yes        No
997                4                1       No        No
998                1                2       No        No
999                2                4       No        No

[1000 rows x 10 columns]
```

In [4]:
```
df.shape
```

Out[4]: (1000, 10)

In [5]:
```
df.dtypes
```

Out[5]:
```
Age                  int64
Gender              object
Department          object
MonthlyIncome        int64
TotalWorkingYears    int64
YearsAtCompany       int64
JobSatisfaction      int64
WorkLifeBalance      int64
OverTime            object
Attrition           object
dtype: object
```

In [6]:
```
df.isnull()
```

Out[6]:

	Age	Gender	Department	MonthlyIncome	TotalWorkingYears	YearsAtCompany	JobSatisfaction	WorkLifeBalance	OverTime	Attri
0	False	False	False	False	False	False	False	False	False	F
1	False	False	False	False	False	False	False	False	False	F
2	False	False	False	False	False	False	False	False	False	F
3	False	False	False	False	False	False	False	False	False	F
4	False	False	False	False	False	False	False	False	False	F
...	...	...	...	...	...	...	...	...	...	
995	False	False	False	False	False	False	False	False	False	F
996	False	False	False	False	False	False	False	False	False	F
997	False	False	False	False	False	False	False	False	False	F
998	False	False	False	False	False	False	False	False	False	F
999	False	False	False	False	False	False	False	False	False	F

1000 rows × 10 columns

In [7]:
```
df.duplicated()
```

Out[7]:
```
0      False
1      False
2      False
3      False
4      False
       ...
995    False
996    False
997    False
998    False
999    False
Length: 1000, dtype: bool
```

In [8]:
```
len(df)
```

Out[8]:
```
1000
```

In [9]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1000 entries, 0 to 999
Data columns (total 10 columns):
 #   Column             Non-Null Count  Dtype 
---  ------             --------------  ----- 
 0   Age                1000 non-null   int64 
 1   Gender             1000 non-null   object
 2   Department         1000 non-null   object
 3   MonthlyIncome      1000 non-null   int64 
 4   TotalWorkingYears  1000 non-null   int64 
 5   YearsAtCompany     1000 non-null   int64 
 6   JobSatisfaction    1000 non-null   int64 
 7   WorkLifeBalance    1000 non-null   int64 
 8   OverTime           1000 non-null   object
 9   Attrition          1000 non-null   object
dtypes: int64(6), object(4)
memory usage: 78.3+ KB
```

In [10]:
```
#how many dept
#display all the department
#avg employee age
#max.min employee age
#what is the gender distribution of employee
```

In [11]:
```
df['Department'].nunique()
```

Out[11]:
```
3
```

In [12]:
```
df['Department'].unique()
```

Out[12]:
```
array(['HR', 'R&D', 'Sales'], dtype=object)
```

In [13]:
```
df['Age'].mean()
```

Out[13]:
```
np.float64(40.947)
```

In [14]:
```
df['Gender'].value_counts()
```

Out[14]:
```
Gender
Male      504
Female    496
Name: count, dtype: int64
```

In [15]:
```
df['Department'].value_counts()
```

Out[15]:
```
Department
HR        352
Sales     343
R&D       305
Name: count, dtype: int64
```

In [16]:
```
df[df["YearsAtCompany"] > 10].shape[0]
```

Out[16]:
```
503
```

In [20]:
```
df[df["YearsAtCompany"] > 5].shape[0]
```

Out[20]:
```
752
```

In [21]:
```
df.groupby("Department")["MonthlyIncome"].mean()
```

Out[21]:
```
Department
HR        11604.042614
R&D       11461.272131
Sales     11417.247813
Name: MonthlyIncome, dtype: float64
```

In [22]:
```
df[df["Attrition"] == "Yes"]["Department"].value_counts()
```

Out[22]:
```
Department
R&D      74
HR       70
Sales    59
Name: count, dtype: int64
```

In [ ]: